

## A Public-Interest Infrastructure for News

# The Newshare Network: Federated Identity, Single Sign-On and Fair Payment for Independent Journalism

*A Proposal for a Missouri Proof-of-Concept Pilot*

In partnership with: Donald W. Reynolds Journalism Institute, University of Missouri • Missouri Press Association

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## The Problem

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Local news is in crisis. The number of local journalists across the United States has declined by more than 75 percent in the past two decades. The economic model that sustained independent journalism — local advertising and subscriptions — has collapsed, largely because a small number of technology platforms have captured the revenue that once supported newsrooms.

The technical root cause is something rarely named plainly: independent publishers have no shared infrastructure for identity, access, or payment. Every reader who wants to support multiple local news organizations must register separately at each one. Publishers cannot recognize a loyal reader arriving from a neighboring publication. No mechanism exists for a reader's home publisher to vouch for them elsewhere, or for content value to be settled fairly across a network of independent publishers — without surrendering everything to a platform intermediary.

*A stalemate among competing interests of marketers, publishers and technology companies is leaving the public with little ability to effectively manage or control personal information coursing through the business or publishing web. There is an urgent need to develop consensus with leadership by a public-benefit entity. — ITEGA, ["Identity, Advertising and the Future of Journalism: A Call to Action," 2021](#)*

The platforms that have stepped into this vacuum — "Sign in with Google," Facebook Connect — are not solutions. They are the problem: surveillance infrastructure that extracts reader data, captures advertiser revenue, and leaves publishers with less every year.

## The Solution: The Newshare Network

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The Newshare Network is a proposed federated identity, single sign-on, and fair-payment network governed by the Information Trust Exchange Governing Association (ITEGA), a California 501(c)3 nonprofit incorporated in 2017. It would allow a reader to create one account at one participating local publisher and be recognized — without re-registering — at any other publisher in the network.

Publishers share users without sharing user data. Transactions are logged to a neutral service and settled periodically through the banking ACH network. The system is built on privacy-by-design principles established in a working prototype built by Clickshare Service Corp. in 1996 and patented in 2008 (U.S. Patent No. 7,324,972, now expired and in the public domain).

## How It Works — For a Reader

- Susan subscribes to the Columbia Missourian. She creates one account there — her “home base.”
- She visits the Joplin Globe, a second network member. She clicks “Log in with your news network account.”
- She is automatically recognized as a Missourian subscriber. She does not re-register.
- The Globe serves her content according to her subscription tier. Her name and email are never transmitted to the Globe.
- At the end of the week, the network settles: the Globe receives a credit, the Missourian a debit, both calculated from anonymous event logs. Susan receives one bill.

## How It Works — For a Publisher

- Publishers set wholesale prices for their content. Home-base publishers set retail prices for their subscribers, capturing the margin as a business opportunity.
- Every publisher gets aggregated usage reports for audit purposes — sorted by home-base ID, never by individual user.
- Publishers retain their direct reader relationships. The network is additive, not a replacement for anything they do today. A publisher accepting the Newshare network is like a merchant accepting Visa: it does not prevent them from issuing their own store card.

## The Privacy Architecture

The Newshare Network is built around a principle established in the 1996 Clickshare prototype: the system works without cookies, without a central identity database, and without any party — including ITEGA — ever having access to a reader’s personal information.

- The authentication and logging service knows users only by an opaque alphanumeric identifier — never by name, email, or any personal data.
- Each reader receives a different anonymous identifier at each publisher they visit, making cross-site tracking technically impossible without the home publisher’s cooperation.
- A reader can “disappear” from a publisher’s records instantly, by their home publisher unlinking the identifier — no action by the publisher required.
- The system is designed to be compliant with GDPR, CCPA, and all current U.S. and European privacy frameworks.

## Track Record

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ITEGA is not a new idea in search of validation. It is a mature concept with a thirty-year technical lineage, a seven-year organizational track record, documented expert consensus, and philanthropic support from Craig Newmark Philanthropies.

### **1994–2008: The Clickshare Proof of Concept**

Bill Densmore founded what became Clickshare Service Corp. in 1994 to solve exactly the problem described above. In 1996, Clickshare deployed a working four-party federated identity and micropayment system serving real publishers and real readers — without cookies, without a central user database, and without a platform intermediary. U.S. Patent No. 7,324,972 was granted in 2008 and expired in 2022. Its architecture is now in the public domain.

### **2008–2009: Reynolds Journalism Institute Fellowship**

Densmore developed the institutional framework for ITEGA as a Donald W. Reynolds Journalism Institute (RJI) Fellow at the University of Missouri in 2008–2009, producing the white paper “From Paper to Persona: A Status Report on the News Ecosystem, and a Challenge to Create the Next One.” RJI has been a consistent partner and supporter since.

### **2017: ITEGA Incorporated**

ITEGA was incorporated as a California public-benefit corporation in January 2017 and received IRS 501(c)3 determination that July. Its restated bylaws were adopted in January 2020. The organization is legally active, financially clean, and governed by a board of directors with no commercial stake in the network’s operation.

### **2018–2021: Craig Newmark-Funded Convenings and Research**

With support from Craig Newmark Philanthropies, ITEGA convened industry stakeholders in Chicago, Brooklyn, and Washington D.C. to build consensus around the need for a public-interest identity and privacy infrastructure for journalism. Three focused webinars in early 2021 brought together dozens of experts in journalism, government, publishing, advertising technology, privacy advocacy, and academic research.

The resulting white paper, [“Identity, Advertising and the Future of Journalism: A Call to Action”](#) (April 2021), documented consensus on four priorities: funding new business models for local news; designating protocols for privacy-respecting user data sharing; creating public auditing and certification mechanisms; and collaborating with governments and industry on digital identity frameworks.

### **2025–2026: Architecture Modernization**

Working with Richard Lerner, Carnegie Mellon-trained computer scientist and CEO of Clickshare Service Corp., ITEGA has updated its technical architecture to modern open standards: OpenID Connect (OIDC), JSON Web Tokens, W3C Verifiable Credentials, and the Really Simple Licensing (RSL) standard for content rights tagging. What required custom C-language wire protocols in 1996 can now be composed from open-source libraries in weeks. A full technical specification was completed in January 2026.

## The Missouri Pilot: Proof of Concept

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ITEGA seeks funding for an 18-month proof-of-concept pilot involving a small consortium of independent Missouri newspapers, hosted at the Donald W. Reynolds Journalism Institute at the University of Missouri. The pilot will demonstrate the core end-to-end network architecture with real publishers and real readers.

### Pilot Partners

| Role                                  | Partner                                                                                                                                                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Institutional Host & Academic Partner | Donald W. Reynolds Journalism Institute (RJI), University of Missouri — RJI has supported ITEGA since its founding and is the natural institutional home for a journalism-industry proof of concept. Bill Densmore is an RJI alumnus fellow (2008–2009). |
| Industry Convener                     | Missouri Press Association (MPA) — will recruit 3–5 participating independent Missouri newspapers (mix of small daily, weekly, and digital outlets).                                                                                                     |
| Technical Operator                    | PubGen.AI – Sho Rust, CEO – developers of a unified CMS platform for local news publishers combining content creation, revenue management and audience analytics into one powerful, AI-driven dashboard. Cape Girardeau, Mo.                             |
| Technical Consultant                  | Clickshare Service Corp. — Richard Lerner, CEO, is co-inventor of the original patent architecture and co-author of the 2026 technical specification. Amherst, Massachusetts.                                                                            |
| Governing Authority                   | ITEGA — sets rules, certifies participants, enforces Exchange Rules.                                                                                                                                                                                     |

### What the Pilot Will Demonstrate

- A reader creates one account at Publisher A (home base) and accesses content at Publisher B without re-registering.
- Publisher B receives the reader's subscription tier information and serves content accordingly.
- The authentication and logging service records the event with an opaque user ID — no personal data transmitted.
- Publisher B cannot identify the reader's real identity from network data.
- Weekly settlement produces accurate usage reports and ACH-ready settlement records.
- At least 50 real readers complete the cross-publisher authentication flow during the pilot.

### What the Pilot Will Not Try to Do

The pilot is a clean-loop demonstration, not a scale deployment. It will not attempt to recruit hundreds of publishers, generate significant revenue, or prove a commercial model. Its purpose is to demonstrate that the architecture works end-to-end with real participants — providing the evidence base for Phase 2 funding and broader rollout.

## Pilot Timeline

| Phase      | Activities                                                                                                                                                                  |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Months 1–2 | Deploy open-source OIDC infrastructure. Configure demo home base. Support or build WordPress publisher plugin. Recruit 3–5 MPA member newspapers. Execute pilot agreements. |
| Months 3–4 | Deploy plugin at participating newspapers. Onboard pilot readers. Launch logging service and weekly settlement report.                                                      |
| Months 5–6 | Live pilot operations with real readers. Monitor, debug, iterate. Collect qualitative feedback from publishers and readers.                                                 |
| Months 7–8 | Expand to additional publishers if pilot is successful. Begin UDEX (User Data Exchange) architecture planning.                                                              |
| Month 18   | Produce public pilot report. Brief funders and potential Phase 2 partners. Prepare Phase 2 funding request.                                                                 |

## Project Budget

ITEGA requests \$350,000 for the 18-month Missouri proof-of-concept pilot. The budget is structured to be lean: it uses open-source infrastructure, leverages existing RJI and MPA relationships, and compensates a small focused technical team.

| Budget Item                                | Amount / Notes                                                                                                                                                                                    |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technical Development                      | \$180,000 Two developers, 12 months: (1) OIDC infrastructure, logging service, settlement engine; (2) WordPress publisher plugin and reader dashboard. PubGen.AI as primary technical contractor. |
| Project Director                           | \$75,000 TBD – Experienced editorial, nonprofit tech and advertising manager and stack developer.                                                                                                 |
| Project Director / Coordination            | \$35,000 Bill Densmore, ITEGA Executive Director, 0.33 FTE for 18 months. Includes publisher outreach, partner management, RJI coordination.                                                      |
| Richard Lerner / Technical Architecture    | \$30,000 CTO/architect role: technical specification, developer oversight, OIDC protocol guidance. Clickshare Service Corp.                                                                       |
| Cloud Infrastructure                       | \$5,400 \$300/month × 18 months. AWS or DigitalOcean hosting for OIDC provider, logging database, settlement service.                                                                             |
| Publisher Integration Support              | \$15,000 Small grants to 3–5 participating newspapers for staff time to integrate the WordPress plugin and onboard readers.                                                                       |
| Legal (Membership Agreements, Pilot Terms) | \$15,000 Draft and finalize pilot membership agreements, exchange rules addendum, and privacy disclosures.                                                                                        |
| Travel and Convenings                      | \$18,000 Missouri site visits, publisher meetings, RJI coordination travel, one mid-pilot stakeholder briefing.                                                                                   |
| Evaluation and Reporting                   | \$12,000 Independent evaluation of pilot metrics; production of public-facing pilot report for funder and industry audience.                                                                      |
| Contingency (10%)                          | \$19,600 For technical unknowns, publisher onboarding delays, or scope adjustments.                                                                                                               |
| TOTAL REQUEST                              | \$400,000                                                                                                                                                                                         |

**Note:** ITEGA is pursuing a consortium funding model. This request may be fulfilled by one funder or shared across two or three. PressForward, the Knight Foundation, the MacArthur Foundation, the Reynolds endowment through RJI, and the Lenfest Institute for Journalism are the primary targets. A matching commitment from RJI (in-kind hosting, staff time, or direct funding) is being sought simultaneously.

## Phase 2 Budget Context

A successful pilot supports a Phase 2 request of approximately \$2 million over three years (consistent with the 2021 OpenNewsSSO project proposal) for broader rollout, UDEX development, and transition to a self-sustaining fee model. Phase 2 is not part of this request.

## Path to Sustainability

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The Newshare Network is designed to become self-sustaining through transaction fees and certification revenue — not perpetual grants. ITEGA's role as governing authority, not operator, is essential to this model: profit flows to publishers and service operators, not to ITEGA. ITEGA's revenue covers only governance, standards maintenance, and reserves.

- Transaction fees: A small percentage of settled value (analogous to credit card interchange, but far lower) paid by licensed ALS operators.
- Annual certification fees: Publishers, home bases, and tech stack providers pay annual fees for ITEGA certification — a revenue stream that can begin before the full network is operational.
- License fees: Commercial operators of authentication, logging, and settlement services pay ITEGA for the right to operate on the network.

The 2021 ITEGA [Call to Action](#) estimated that network infrastructure and startup costs borne by ITEGA could be less than \$2 million, with operators financing their own technical investment and paying network fees. “The thing is if you get this up and going one could actually turn to venture capital firms to expand the market once the idea is well put together,” said Robert Picard of the Reuters Institute at Oxford University.

## Why 2026 Is the Right Moment

| Factor                       | Why It Matters Now                                                                                                                                                                                                                                                        |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standards exist              | OpenID Connect, OAuth 2.0, JWT, and W3C Verifiable Credentials provide a modern protocol foundation that maps directly onto the 1996 Clickshare architecture. The build is now weeks, not years.                                                                          |
| Third-party cookies are dead | Google's deprecation of third-party cookies and Apple's App Tracking Transparency have eliminated the surveillance infrastructure publishers once depended on for ad targeting. The ITEGA model is the privacy-respecting replacement.                                    |
| Regulatory tailwind          | GDPR, CCPA, and anticipated federal privacy legislation create both the mandate and the market incentive for privacy-by-design identity infrastructure. Publishers who build this now will be ahead of compliance requirements.                                           |
| RSL content tagging          | The <a href="#">Really Simple Licensing</a> (RSL) standard (2025) provides a uniform protocol for tagging content ownership and usage rights across the web, including for AI model training. ITEGA's network is positioned to be the first journalism-sector deployment. |
| AI content licensing urgency | As AI companies negotiate bulk content licensing deals with publishers, the lack of a neutral infrastructure for metering, logging, and settling these transactions is an acute gap. The ITEGA architecture solves this at the network level.                             |
| Platform fragility           | The dominance of Google and Meta is increasingly fragile — regulatory, legal, and reputational pressure is mounting. Publishers need an alternative that does not require a platform intermediary. There is no other nonprofit-governed candidate.                        |

## Leadership

| Person                                                      | Background                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bill Densmore Founder & ITEGA Executive Director, (interim) | Career journalist and tech entrepreneur. AP editor/writer in Boston, Chicago, and San Francisco. Co-owner/publisher, The Advocate newswEEKlies (Berkshires), 1983–1992. Founded Clickshare Service Corp., 1994. RJI Fellow, 2008–2009. Author, <a href="#">“From Persona to Payment: A Status Report on the News Ecosystem, and a Challenge to Create the Next One,”</a> (2015). B.A., University of Massachusetts Amherst (environmental policy and communications). Based in Williamstown, Mass. |
| Richard Lerner CEO, Clickshare Service Corp.                | Carnegie Mellon University Ph.D. in computer science. Co-inventor, U.S. Patent No. 7,324,972. Co-author, ITEGA SSO Network Description (2026). Lead technical architect of the Newshare Network reference implementation. Based in Amherst, Mass.                                                                                                                                                                                                                                                  |
| Jo Ellen Green Kaiser, Board Chair, ITEGA                   | Non-executive lead of ITEGA's governance and strategic direction. Moderated the 2021 webinar series that produced the <a href="#">ITEGA Call to Action</a> .                                                                                                                                                                                                                                                                                                                                       |



## Organizational Assets

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ITEGA brings a suite of assets that would take years and significant capital to replicate:

- 501(c)3 status since 2017, restated bylaws 2020, clean financial history.
- Expired patent (public domain) architecture with 30-year lineage — the foundational technical design is proven, not experimental.
- Domain portfolio: itega.org, infotrust.org, newshare.com (service mark), newssso.org, newssso.com, udex.org.
- Complete governance documents: Exchange Rules (v3.1h), Member Services Agreement (v2.0d), Governance Goals & Structure (v3.0), Technical Design Specification (v6, Jan. 2026), SSO Network Description (v4, Jan. 2026).
- Documented expert consensus from Chicago, Brooklyn, D.C. convenings and three 2021 webinars — funded by Craig Newmark Philanthropies.
- Established relationships with RJJ, Missouri Press Association, and the Local Media Consortium.
- Active website (itega.org), Privacy Beat newsletter, social media presence.

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### The Ask

**\$400,000 over 18 months**

*To fund a Missouri proof-of-concept pilot demonstrating the end-to-end Newshare Network architecture*

*with real publishers, real readers, and real settlement —*

*governed by a public-benefit nonprofit with no commercial stake in the outcome.*

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| Contact              | Bill Densmore, Executive Director, (interim), Information Trust Exchange Governing Association 75 Water Street, PO Box 367, Williamstown, MA 01267 wpdensmore@itega.org • 617-448-6600 itega.org                                                                                                    |
| Supporting Documents | Full technical specification: “Newshare Network: Technical Architecture and Specification” (Feb. 2026) ITEGA Call to Action white paper (April 2021): itega.org/itega-calls ITEGA Exchange Rules, Member Agreement, Governance Structure (Jan. 2026) Clickshare patent (public domain): US7324972B1 |

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